Improving On-site Reservoir Operation Policies by Robust and Adaptive Optimization Models

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<Improving On-site Reservoir Operation Policies by Robust and Adaptive Optimization Models>

C4. Coupling Hedging Rules with Optimization Models in Reservoir Operation Problems (Kim & Kim 2021)

C5. Reforming the Hedging Rules with Adaptive Operations against Multi-year Droughts (*currently in press*)

C6. Improving the Robustness of Reservoir Operations with Stochastic Dynamic Programming (Kim et al. 2021)

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How Does the Coupling of Real-World Policies with Optimization Models Expand the Practicality of Solutions in Reservoir Operation Problems?

Gi Joo Kim[1] · Young-Oh Kim[1]

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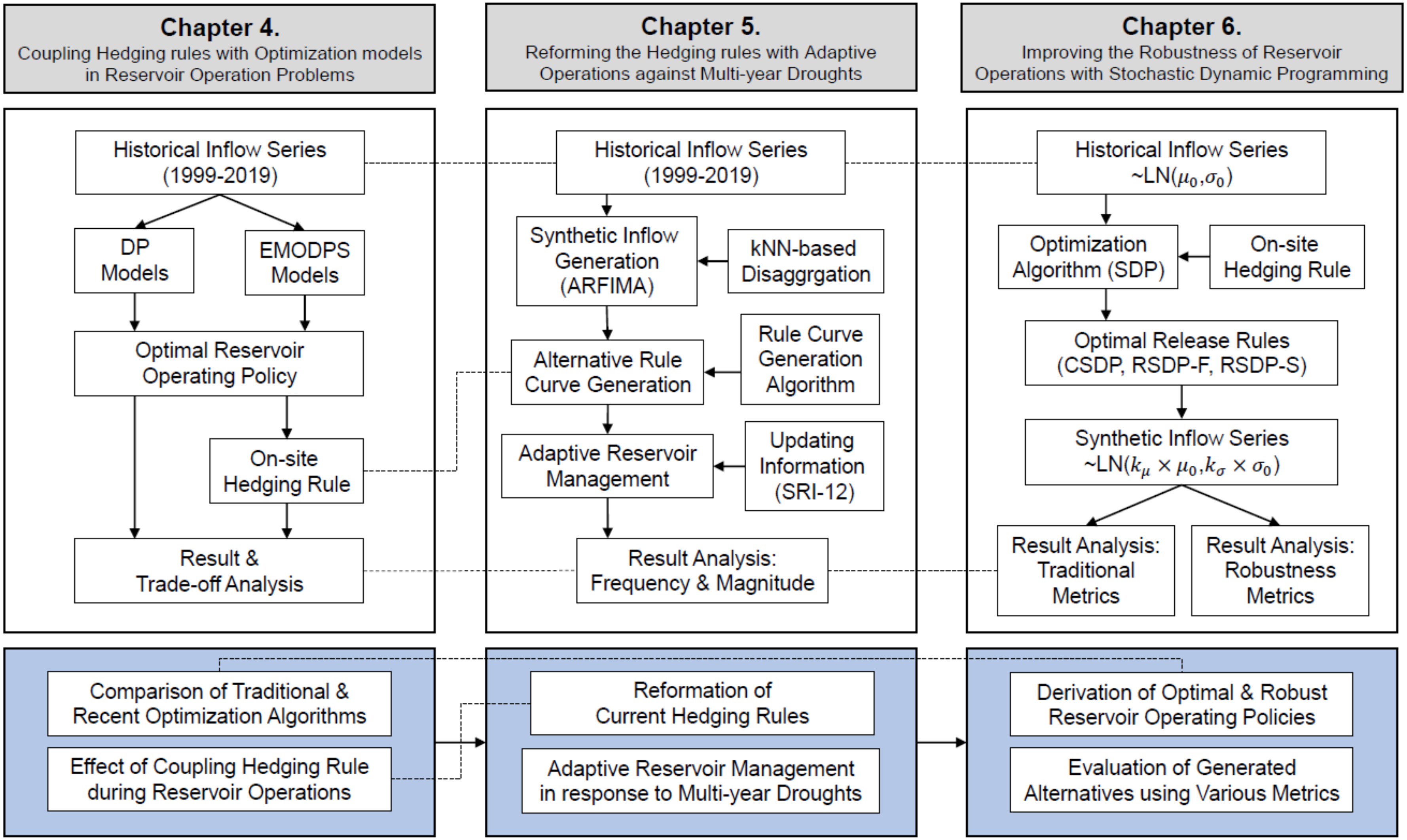
Improving the Robustness of Reservoir Operations with Stochastic Dynamic Programming

Gi Joo Kim[1]; Young-Oh Kim[2]; and Patrick M. Reed, M.ASCE[3]

Abstract: Reservoir operations should consider both adaptiveness and robustness to deal with two of the main characteristics of climate change: nonstationarity and deep uncertainty. In particular, robust operational strategies are distinguished from risk-neutral expected value optimization in the sense that they should be satisfactory over a wider range of uncertainty and improve the ability of a reservoir system to adapt to climate change. In this study, a new framework named robust stochastic dynamic programming (RSDP) is proposed that couples robust optimization (RO) with the formulations of objective function or constraints used in stochastic dynamic programming (SDP). Two main approaches of RO, namely feasibility robustness and solution robustness, are both considered in the optimization algorithm. Consequently, this study uses the Boryeong multipurpose dam to evaluate three SDP framings: conventional-SDP (CSDP), RSDP-feasibility robustness (RSDP-F), and RSDP-solution robustness (RSDP-S). These three SDP formulations were used to derive optimal monthly release rules for the Boryeong Dam, and their relative performances were evaluated using simulations of a broader range of inflow scenarios. The simulation-based re-evaluations of the resulting reservoir operational policies were quantified using a wide range of metrics that include reliability, resiliency, and vulnerability, as well as regret-based robustness metrics. The results of this study suggest that the RSDP-S model not only increases the range of possible solutions, but also yields more desirable operation outcomes under extreme climate conditions with respect to both traditional and robustness metrics. **DOI:** 10.1061/(ASCE)WR.1943-5452.0001381. *© 2021 American Society of Civil Engineers.*

Author keywords: Reservoir operation; Stochastic dynamic programming; Robust optimization.



Chapter 4.
Coupling Hedging rules with Optimization models in Reservoir Operation Problems
Historical Inflow Series (1999-2019)
DP Models
EMODPS Models
Optimal Reservoir Operating Policy
On-site Hedging Rule
Result & Trade-off Analysis
Comparison of Traditional & Recent Optimization Algorithms
Effect of Coupling Hedging Rule during Reservoir Operations
Chapter 5.
Reforming the Hedging rules with Adaptive Operations against Multi-year Droughts
Historical Inflow Series (1999-2019)
Synthetic Inflow Generation (ARFIMA)
kNN-based Disaggrgation
Alternative Rule Curve Generation
Rule Curve Generation Algorithm
Adaptive Reservoir Management
Updating Information (SRI-12)
Result Analysis: Frequency & Magnitude
Reformation of Current Hedging Rules
Adaptive Reservoir Management in response to Multi-year Droughts
Chapter 6.
Improving the Robustness of Reservoir Operations with Stochastic Dynamic Programming
Historical Inflow Series ~LN(μ_0,σ_0)
Optimization Algorithm (SDP)
On-site Hedging Rule
Optimal Release Rules (CSDP, RSDP-F, RSDP-S)
Synthetic Inflow Series ~LN($k_\mu \times \mu_0$,$k_\sigma \times \sigma_0$)
Result Analysis: Traditional Metrics
Result Analysis: Robustness Metrics
Derivation of Optimal & Robust Reservoir Operating Policies
Evaluation of Generated Alternatives using Various Metrics

C4. Coupling Hedging Rules with Optimization Models in Reservoir Operation

- ❖ How does **different optimization models or their formulations** affect both the **quality and the diversity** of solutions in reservoir operation problems?

- ❖ How about when the reservoir was operated with the **status quo (SQ)** rules? How do they perform under both drought and normal conditions compared to optimal solutions?

- ❖ How does the **implementation of SQ rules with the solutions from optimization algorithms** improve (or degrade) the performance under extreme drought conditions?

C4. Coupling Hedging Rules with Optimization Models in Reservoir Operation

❖ 2-Obj. Optimization Models for Optimal Reservoir Operations

◆ Dynamic Programming-based Models with different information

- Deterministic DP-Perfect (DDP-Perfect)
- Deterministic DP-Monthly Average (DDP-Average)
- Stochastic DP (SDP): Assuming lognormal distribution of inflows

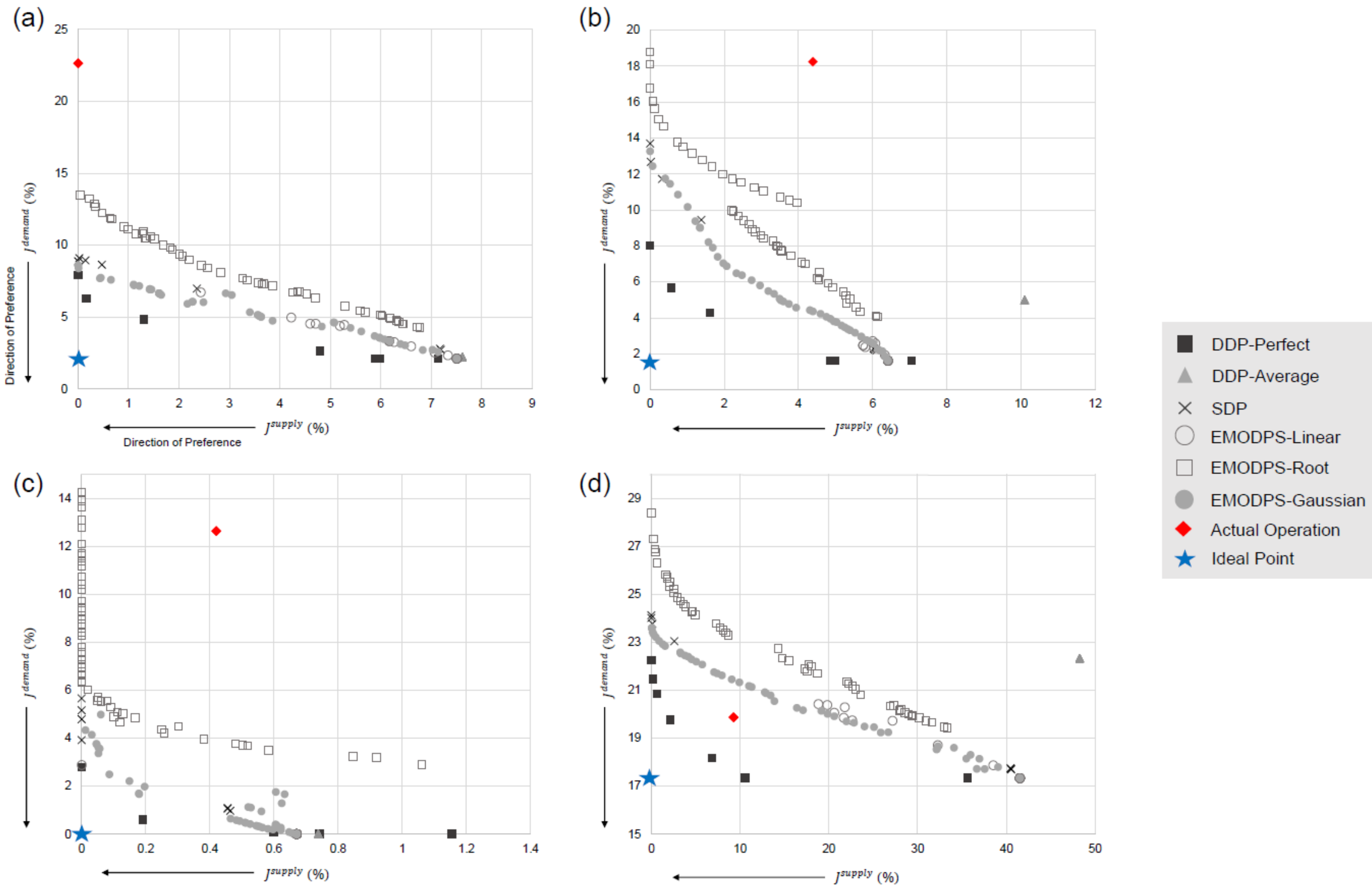
$$f^{n}{}_{opt}(S_t, Q_t) = \max_{R^*_t}\left[Z_t(S_t, Q_t, R_t) + \underset{Q_{t+1}|Q_t}{E} f^{n-1}{}_{opt}(S_{t+1}, Q_{t+1}) \right] \qquad Z_t(\bullet) = w_1 \times J_t^{supply} + w_2 \times J_t^{demand}$$

◆ Evolutionary Multi-Objective Direct Policy Search (EMODPS) Models with different RBF shapes & number of parameters

- EMODPS-Linear: Linearly shaped RBF with 4 policy parameters
- EMODPS-Gaussian: Gaussian shaped RBF with 3 policy parameters
- **EMODPS-Root**: Root shaped RBF with 6 policy parameters (j=2)

$$R_t = \min\left(\max\left(\sum_{i=1}^{j} \left(w_i \times \left| D_t - \frac{S_t - d_i}{r_i} \right|^{0.5} \right), R_{\max} \right), R_{\min} \right)$$

C4. Coupling Hedging Rules with Optimization Models in Reservoir Operation



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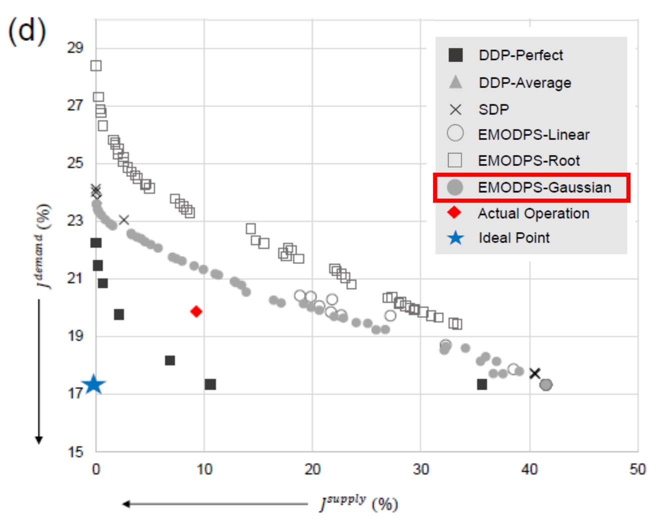
C4. Coupling Hedging Rules with Optimization Models in Reservoir Operation

❖ During simulation, real-world reservoir operating policy is coupled with the best performing optimization model: **EMODPS-Gaussian**



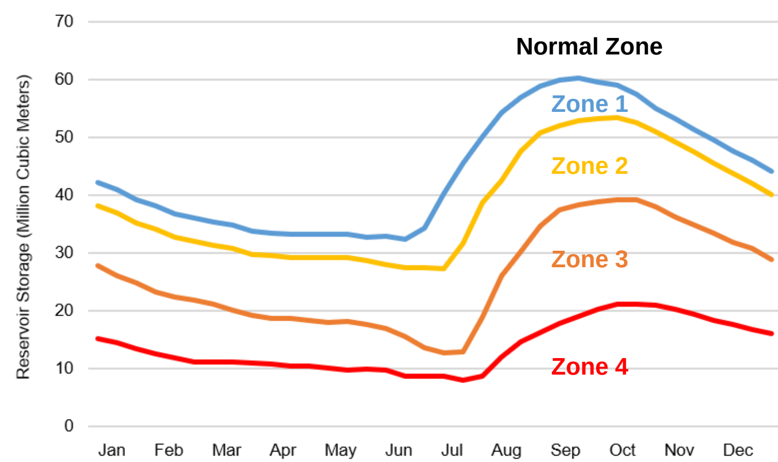


Fig. 9. Actual Rule Curves of Boryeong Reservoir, Korea

Table 3. Ratios implemented during hedging

Zone Name	Municipal & Industrial	Irrigation	Environmental
Normal Zone & Zone 1 (Attention Zone)	100%*	100%*	100%*
Zone 2 (Caution Zone)	100%	100%	Up to 0%
Zone 3 (Alert Zone)	100%	Up to 70%	0%
Zone 4 (Critical Zone)	Up to 80%	70%	0%

* Release exceeding demand is possible in Normal Zones

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C4. Coupling Hedging Rules with Optimization Models in Reservoir Operation

❖ Selected Performance Indices

◆ Risk, Resiliency, and vulnerability (Hashimoto et al. 1982)

- Risk (frequency): How frequent was the failure?
- Resiliency (duration): How fast does the system come back from failure?
- Vulnerability (magnitude): How big is the magnitude of the failure?

◆ Evaluation of performance both from the Supply & Demand side (Definition of Failures)

- Supply side (= J^{supply}): $S_t < S_t^{target}$
- Demand side: $R_t < D_t^{M\&I}$ or $R_t < D_t^{Agr}$

C4. Coupling Hedging Rules with Optimization Models in Reservoir Operation

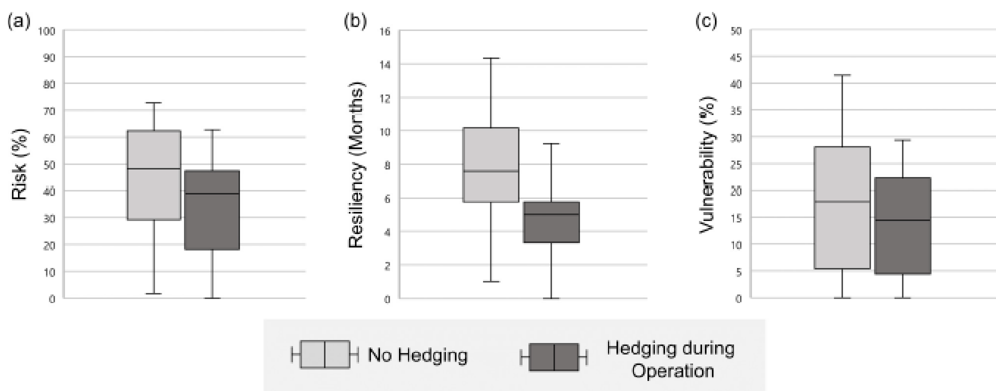


Fig. 10. Performance indices measured from the supply side: (a) risk, (b) resiliency, and (c) vulnerability

Risk Aversion

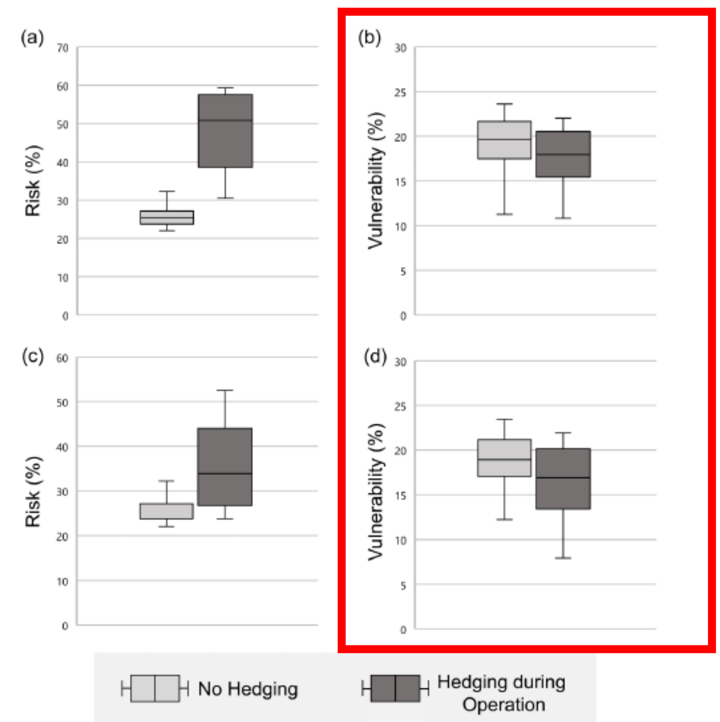
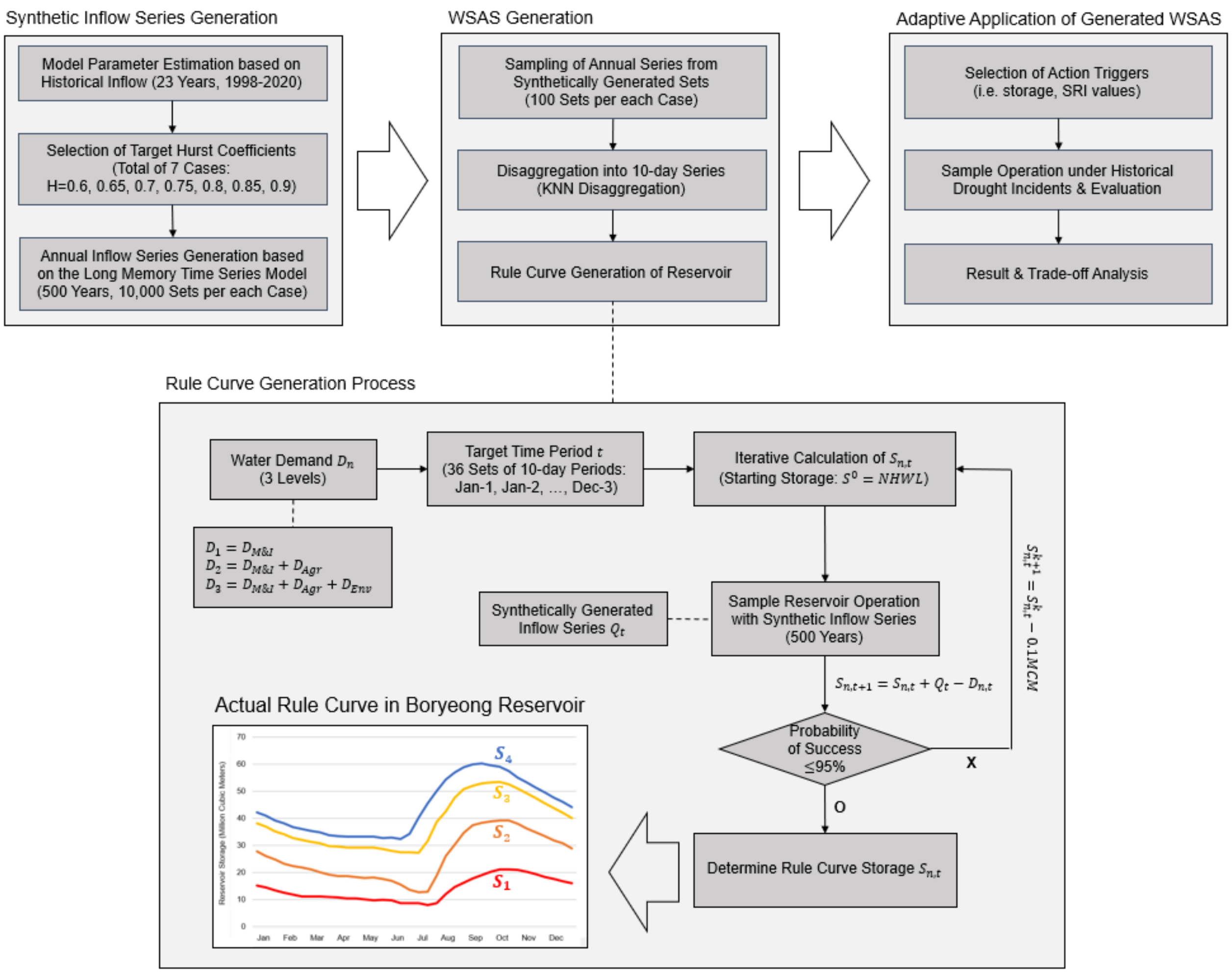


Fig. 11. Performance indices measured from the demand side: (a) risk of water for A uses, (b) vulnerability of water for A uses, (c) risk of water for M&I uses, (d) vulnerability of water for M&I uses,

❖ Coupling of zone-based hedging rule with the solutions from the optimization models resulted in outcomes favorable to risk-averse real-world water users (may hold promise)

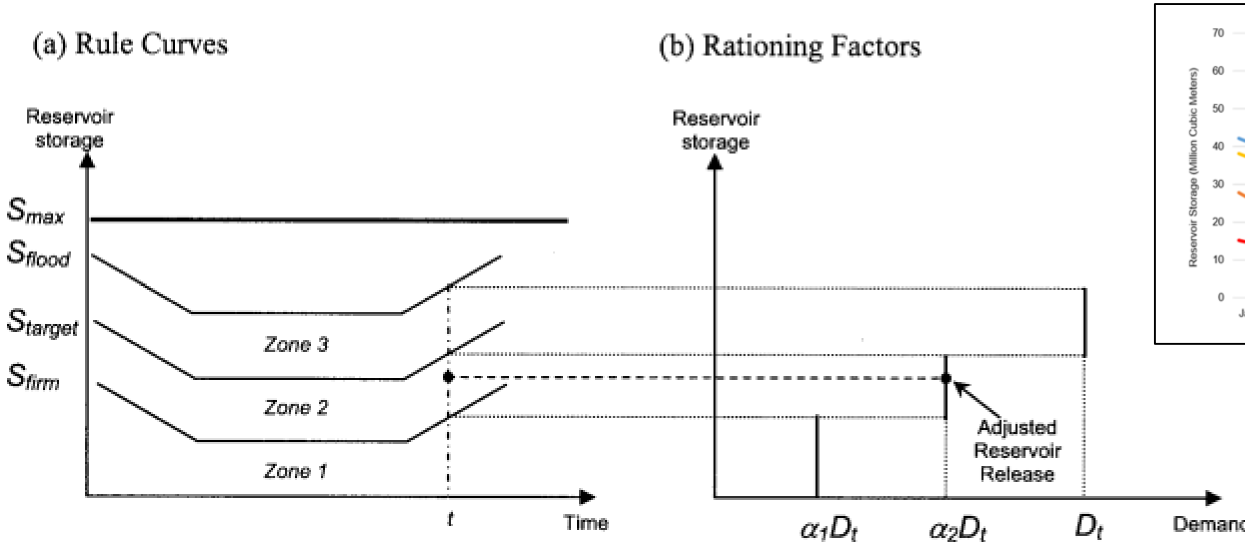
C5. Reforming the Hedging Rules with Adaptive Operations against Multi-year Droughts



❖ Keywords: Zone-based hedging rule, K-nearest neighbor disaggregation, Adaptive reservoir operations, Multi-year drought

Methodology: Hedging Rules

- ❖ Standard operating policy (SOP): release water as close to the target demand as possible (saves only surplus water for future delivery)
- ❖ Hedging (rationing operation) for reservoir operations
  - ◆ Provides only a portion of the target when greater release is possible
  - ◆ Types of hedging rules: One-point, two-point, discrete (zone-based), …



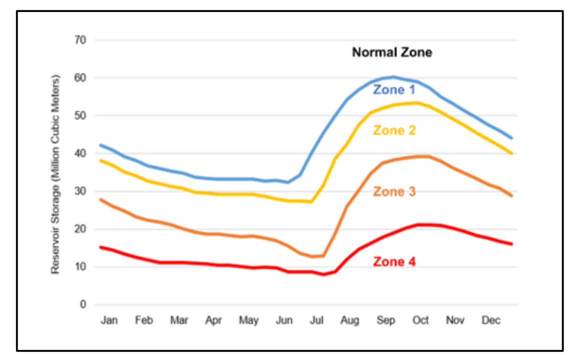


Fig. 1. Concept of zone-based hedging in reservoir operations, adopted from Tu et al. (2008)

Methodology: Long Memory Time Series Generation

❖ Hurst Phenomenon (Hurst, 1951)

- Long-range dependence (LRD), long memory, long term persistence
- Hurst Coefficient H: $(0 < H < 1)$
  - Measure of long-term memory of time series (long term persistence)
  - Series with H values greater than 0.5 indicates **long memory**

❖ Autoregressive Fractionally Integrated Moving Average model (ARFIMA)

$d = H - 1/2$

- ARFIMA(p,q,d): $\underbrace{\Phi(B)}_{\text{AR Term}}(1-B)^{d} X_t = \underbrace{\Psi(B)}_{\text{MA Term}}\varepsilon_t$

  B: Backshift Operator

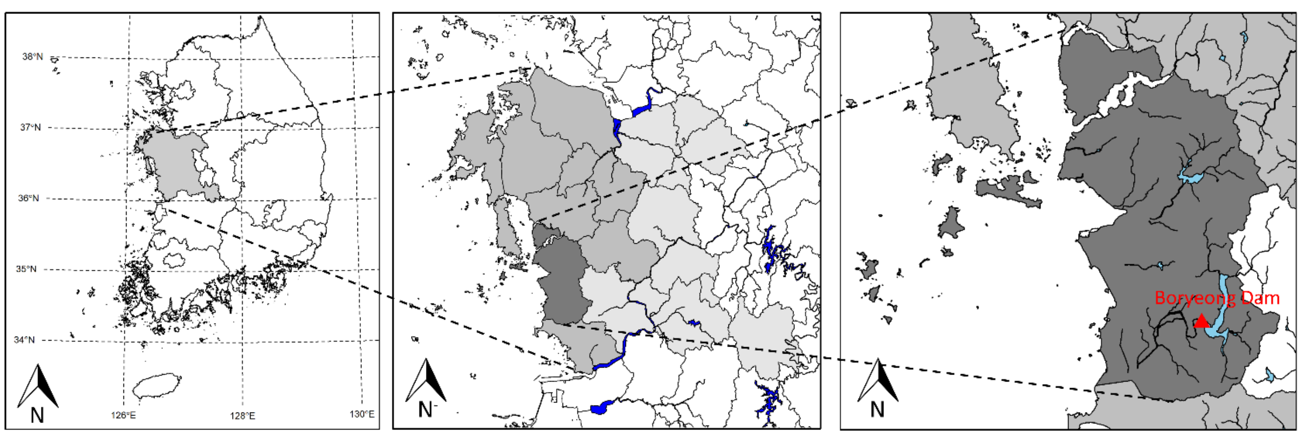
- Ables to capture the long term persistence in the time series: series with d greater than 0 (i.e., H>0.5) indicates series with long term positive autocorrelation: consequently to multi-year floods/droughts

Case Study: Boryeong Multipurpose Reservoir

❖ Watershed Area: 2,860 km^2

❖ Water Supply Districts (8, Light Gray)
Boryeong (Dark Gray), Seosan, Dangjin, Seocheon, Cheongyang, Hongseong, Yesan, Taean

❖ Water Use: 106.6 MCM/year (Reservoir Capacity: 116.9 MCM)
(**Municipal & Industrial water demand: 84%,**
Agricultural water demand: 5%, Environmental water demand: 11%)



Republic of Korea | South Chungcheong Province | Boryeong City

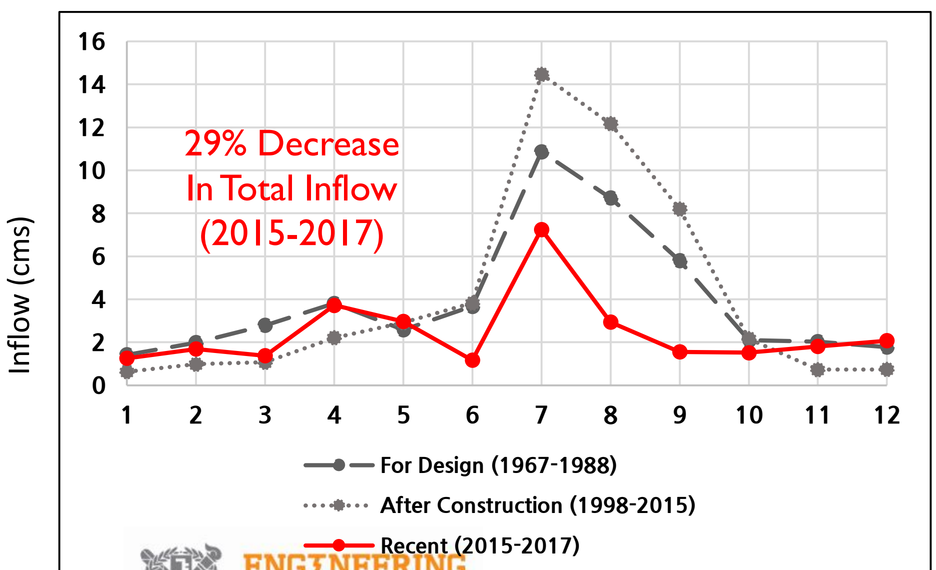
Case Study: Boryeong Multipurpose Reservoir

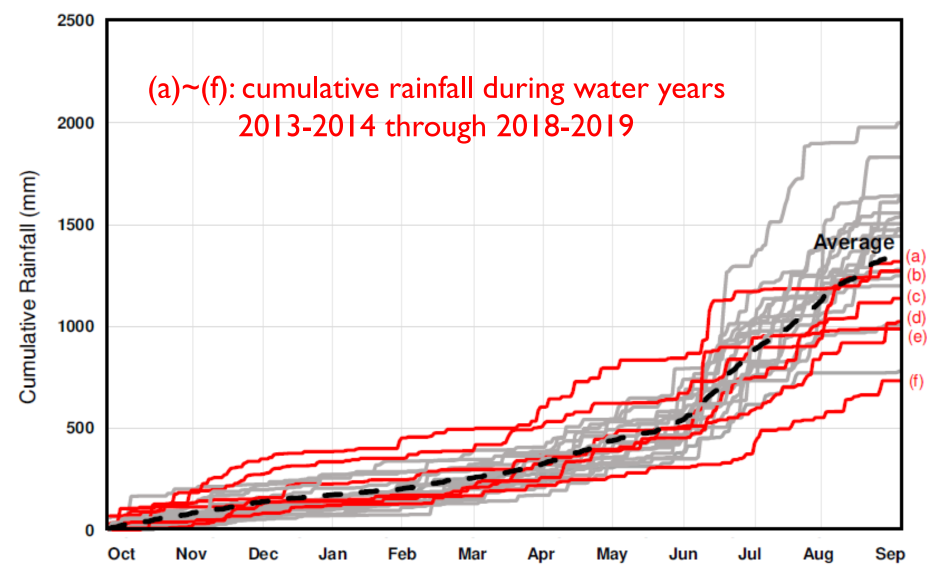
❖ Recent Multi-year Drought in Boryeong Dam Basin (2014-2019)

◆ Consecutive decrease in the total annual precipitation (6 water years)

◆ Both structural/non-structural measures to mitigate the effects from the long-lasting drought

- Conduit Construction Project (2016-2017, $62,500,000+$2,600,000/year)
- Enforcement of Pressure Reduction in Water Supply Systems (127 days)





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Application: SQ Rule Generation Algorithm

❖ Inclusion of Long Memory Time Series during rule curve generation

- ◆ Statistics of synthetically generated inflow series $Q_{t,H}$
  - Average, standard deviations: Identical to historical statistics
  - 7 Cases of Long persistence (**H** = 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9)

Stronger Long Memory

Rule Curve Generation Process

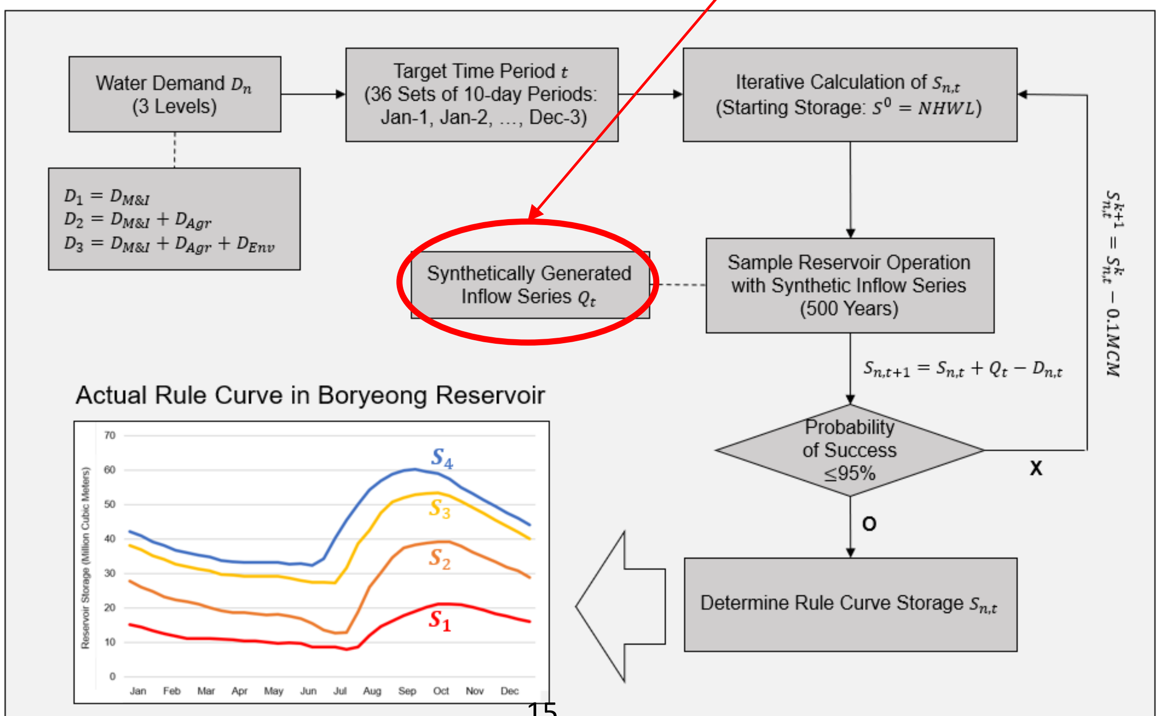


Fig. 2. Iterative calculation process of determining the storages in each zones

Results: Synthetic Inflow Generation

❖ Generation of Annual Series with different long term persistence

❖ Disaggregation from Annual series to 10-day series

◆ Utilization of K-nearest neighbor (KNN) Disaggregation (Nowak et al. 2010)

◆ Preserves the distributional statistics (i.e., mean, variance, skewness, and minimum & maximum values) of the historical records

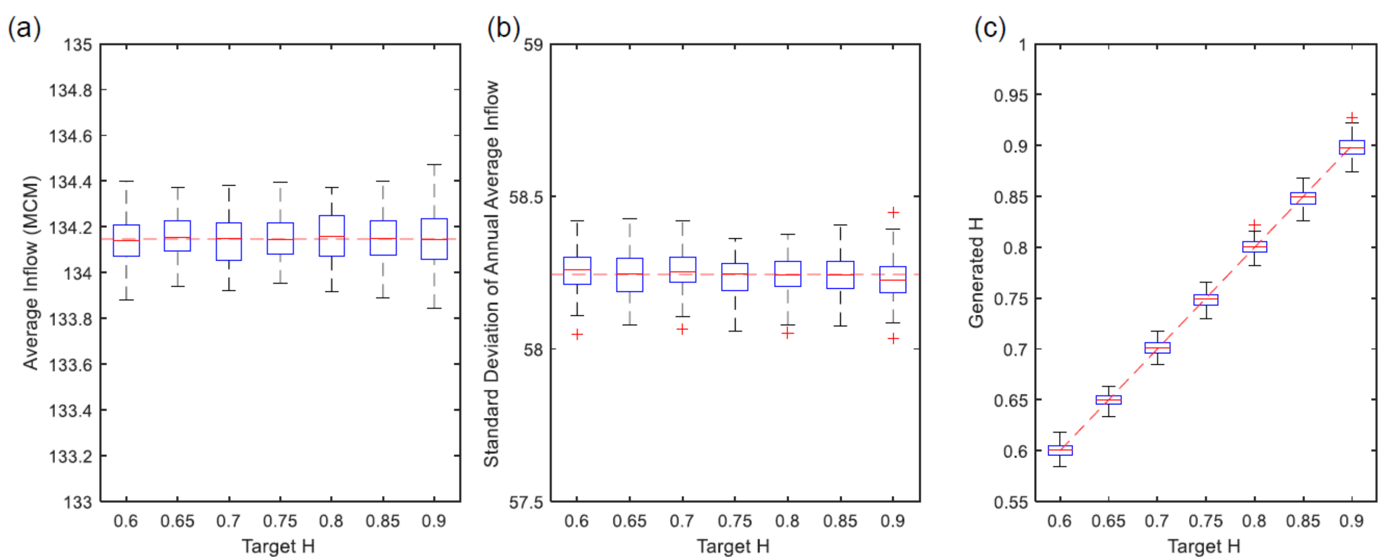
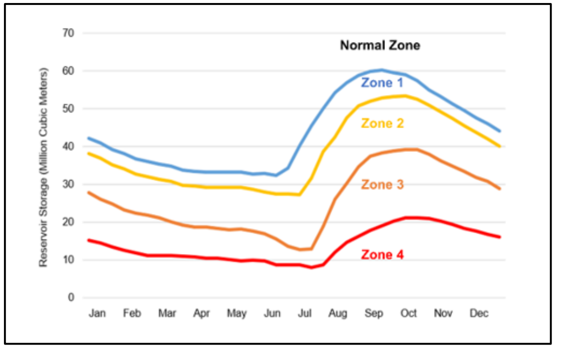


Fig. 3. Statistics of the generated annual inflow series with long memory: (a) average, (b) standard deviation, and (c) Hurst coefficient

Results: SQ Rule Formulations



❖ Iterative calculations for rule curve generation

- 100 Sets of 500-year series for more robust rule curve storage values
- Fig. 4 shows 3 representative cases where darker = stronger long term persistence (H=0.6, 0.75, 0.9)

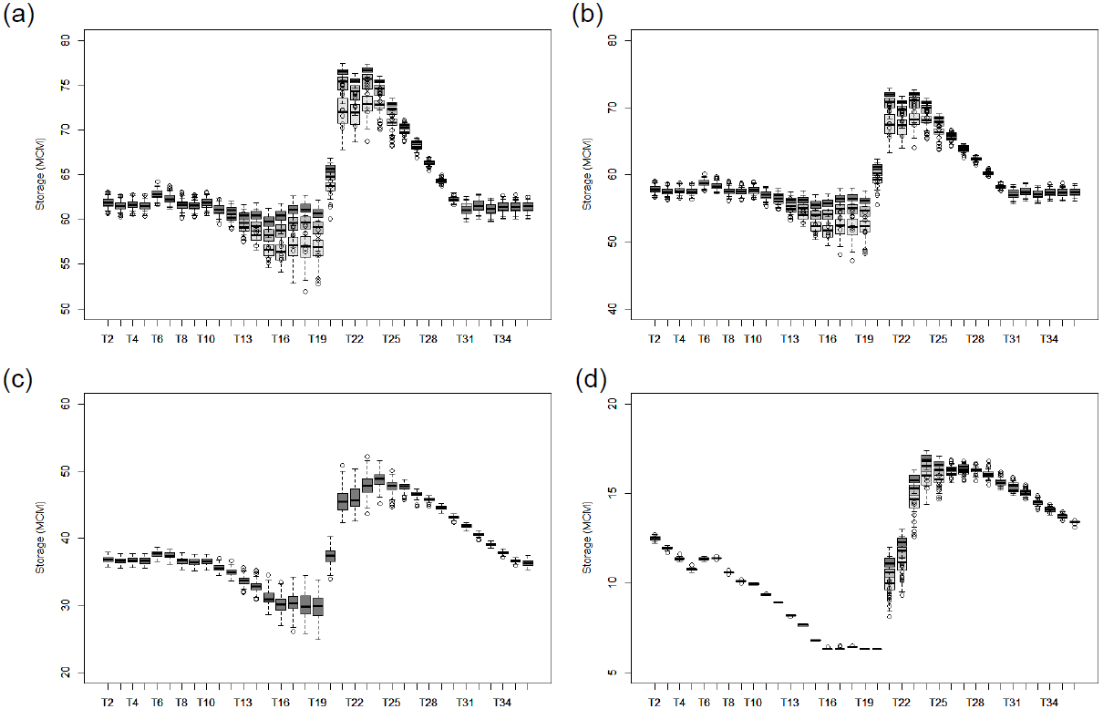


Fig. 4. Generated alternative storage zones:
(a) zone 1 (b) zone 2 (c) zone 3 (d) zone 4 (attention-caution-alert-critical)

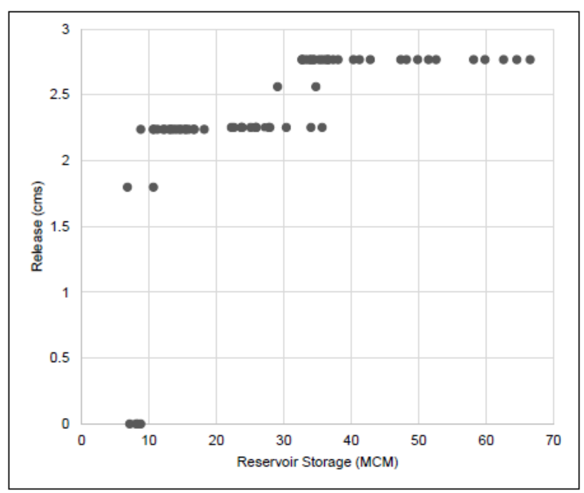
Results: Adaptive RO with Alternative Rules

❖ Adaptive Reservoir Operations with Alternative Rule Curves

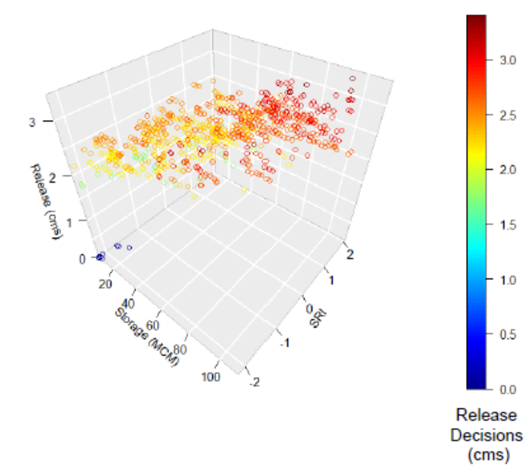
- **Adaptive (or dynamic) DM** = decisions are iteratively updated by the results of its previous outcomes and new information (Kolokytha et al. 2017)
- Selected Trigger: Standard Runoff Index (SRI) 12

SRI Trigger Value	Cumulative Probability	Interval Probability	Utilized Policy
0 <			
−0.180			
−0.366 <			
−0.566 <			
−0.792 <			
−1.068 <			
−1.465 <			
SRI ≤ −1.465	7.14%	7.14%	Case 7 (H = 0.9)

(a)



(b)



Results: Adaptive RO with Alternative Rules

❖ Reservoir Simulation Comparison (under historical inflow, 1998-2020)
K-water SQ Rule vs. Adaptive Operations (proposed in this research)

- Supply (a): Storage (%), darker columns indicate lower storage
- Demand (b): Water Deficit Ratio (%), darker columns indicate greater water deficit ratio

Critical Failures (>30%):
SQ 11.35% vs AO 2.78%

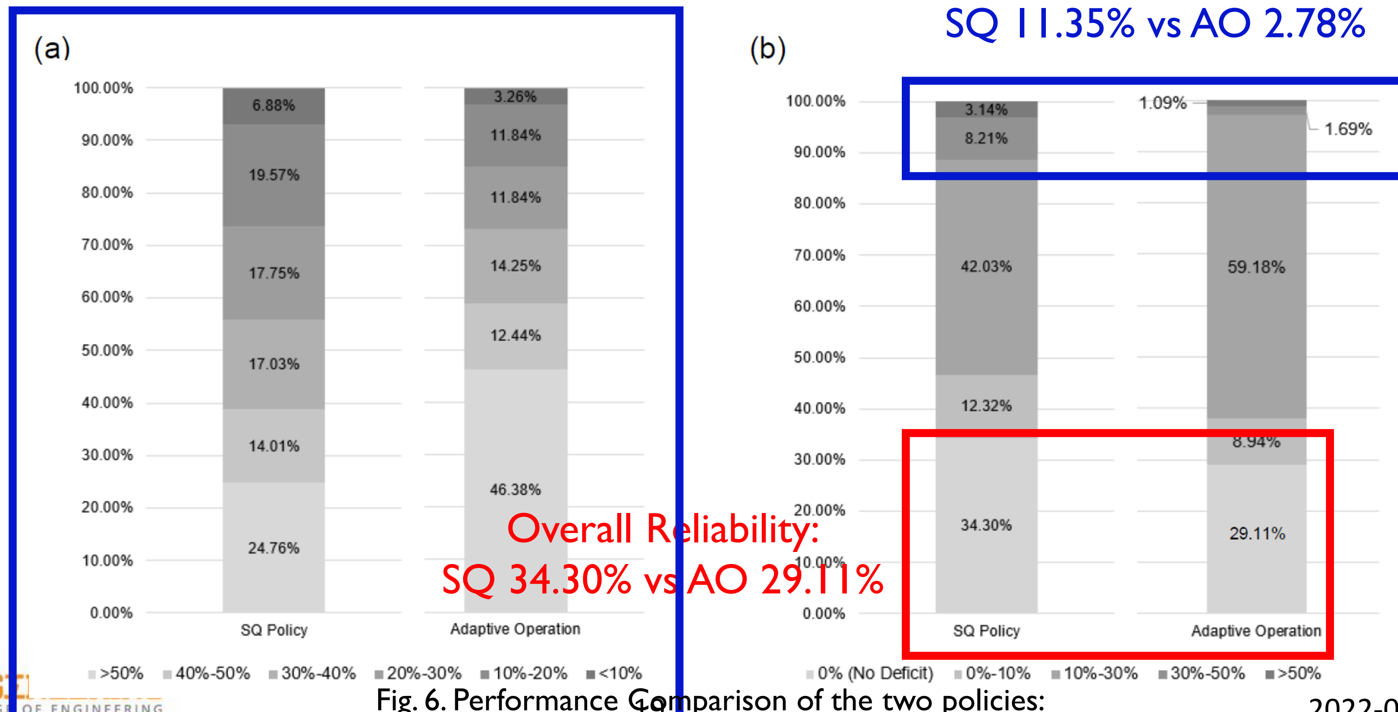


Fig. 6. Performance Comparison of the two policies:
(a) supply side performance (b) demand side performance

Summary & Conclusion

❖ Reformed SQ Policy (Alternative rule curves)

- ◆ Variability of the resulting policy mostly during the flooding seasons (T13-T25, May-01 through Sep-10)
- ◆ Stronger long memory inflow series leads to higher (or more conservative rule curve values, especially in zones 1, 2 & 4)

❖ Adaptive RO with Alternative Hedging Rules

- ◆ Adaptive application of alternative rule curves based on the updating information: SRI-12 values to account for long-term droughts
  - Enables more flexible release decisions
- ◆ Adaptive operations simulations under historical droughts decreased the frequency of critical failures
- ◆ Can be utilized especially under extreme drought conditions

C6. Improving the Robustness of Reservoir Operations with Stochastic Dynamic Programming

❖ Stochastic Dynamic Programming

$$f_t(S_t, H_t) = \min_{R_t}\{Z_t(S_t, Q_t, R_t) + E_{H_{t+1}|H_t, Q_t}[\, f_{t+1}(S_{t+1}, H_{t+1})]\,]\}$$

: Recursive Equation

$$S_{t+1} = S_t + Q_t - R_t$$: Continuity Equation

$$R_t = \min\{\max[R_t, S_t + Q_t - S_{max}], S_t + Q_t - S_{min}\}$$: Feasibility Constraint

Z_t : Objective function, R_t : Release, Q_t : Inflow, H_t : Hydrologic state variable

S_t^k : Storage at time t, S_{min} & S_{max} : Minimum/Maximum storage

❖ Robust Optimization (Mulvey et al. 1995)

$$\underset{\mathbf{x},\mathbf{y}}{Minimize}\ \boxed{\sigma(\mathbf{z}_1, \ldots, \mathbf{z}_S) + \omega p(\alpha_1, \ldots, \alpha_S)}\ ^{*}$$

: Robustness Objective Function

Subject to:

$$\mathbf{Ay} = \mathbf{b}$$

$$\mathbf{g}(\mathbf{x}_s) + \mathbf{B}_s\mathbf{y} + \boldsymbol{\alpha}_s = \mathbf{d}_s$$

$$\mathbf{x}_s, \mathbf{y} \geq 0,\ s \in \Omega$$

Measure of how much the control constraint is violated in RO

C6. Improving the Robustness of Reservoir Operations with Stochastic Dynamic Programming

$$\underset{\mathbf{x},\mathbf{y}}{Minimize}\ \boxed{\sigma(\mathbf{z}_1,\ldots,\mathbf{z}_S)+\omega p(\alpha_1,\ldots,\alpha_S)}$$

: Robustness Objective Function : Robust Feasibility Constraint

$$R_t=\min\{\max[R_t,S_t+Q_t-S_{\max}],S_t+Q_t-\boxed{S_{\min,R}}\}$$

❖ Two concepts of Robustness

◆ Feasibility robustness:

$$\mathbf{g}(\mathbf{x}_s)+\mathbf{B}_s\mathbf{y}+\boldsymbol{\alpha}_s=\mathbf{d}_s$$

range of search for a potential solution is expanded by the use of soft constraints and corresponding penalty term

◆ Solution robustness

the variance of the initial objective is penalized according to the distance from either its best value or its most frequent value

$$\boxed{\max_S \mathbf{z}_s} \qquad \boxed{\underset{z_S}{\arg}\max_S p(\mathbf{z}_s)}$$

❖ Couples key components of RO into the objective function and constraints of SDP (Both feasibility and solution robustness)

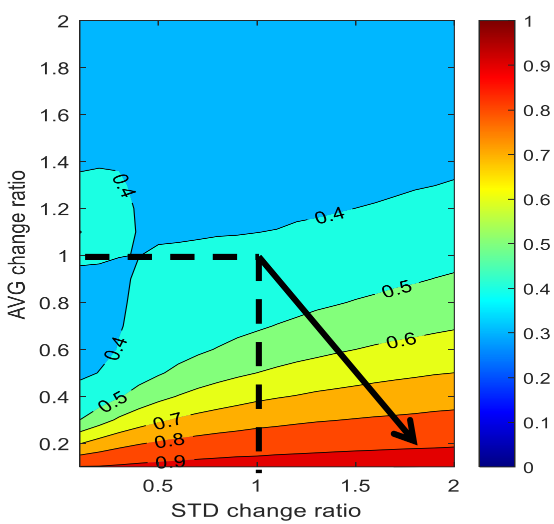
C6. Improving the Robustness of Reservoir Operations with Stochastic Dynamic Programming

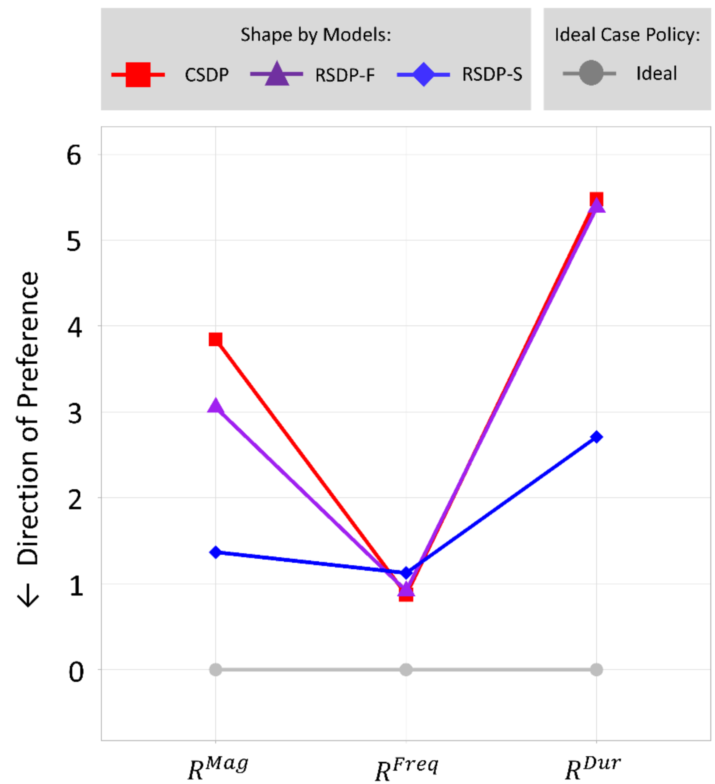
❖ Regret-based Robustness Metric: R^i_{model}

$$R^i_{model} = [D_{i,90} : P(D_i \le D_{i,90}) = 0.90]$$

- ◆ Measures the 90th percentile deviation from the average performance using the baseline scenario

$$D_{i,j} = \frac{|F(\bar{\mathbf{x}})_{i,j} - F(\bar{\mathbf{x}})^*_i|}{F(\bar{\mathbf{x}})^*_i}$$





Ongoing Research Topics

❖ Improvement in probabilistic drought prediction method reflecting drought-related information

◆ Development of hydrological drought prediction method with ensemble drought prediction (EDP)

- Precipitation forecast + soil moisture -> update the prior EDP (Bayesian)
- Results of the Bayesian update were named as EDP+A and EDP+AS
- Performance evaluated with RMSE, ranked probability skill score (RPSS), Brier skill score (BSS)

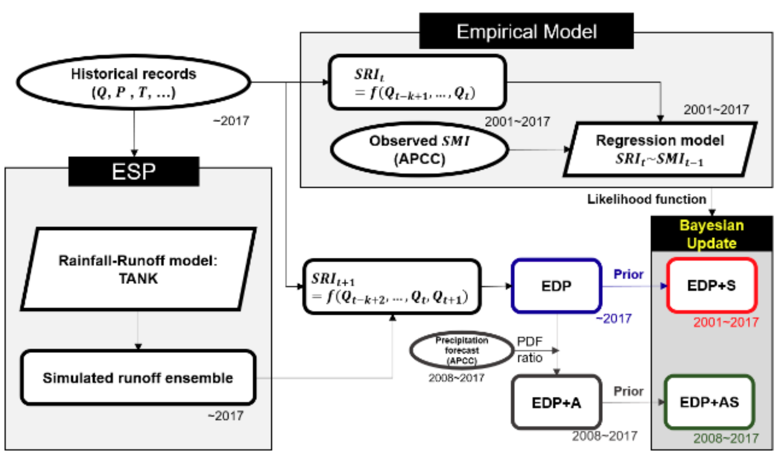


FIG. 4. Modeling framework

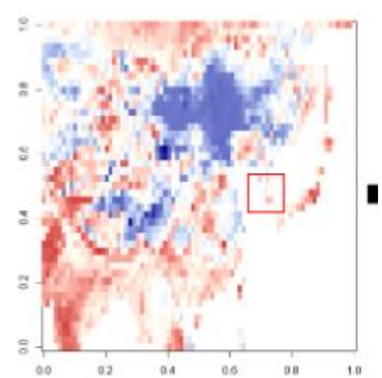
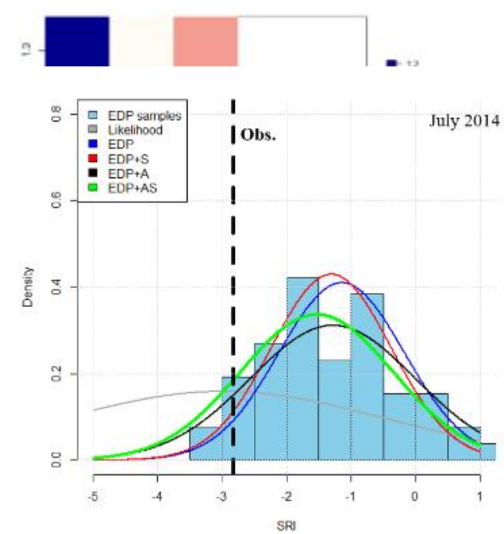
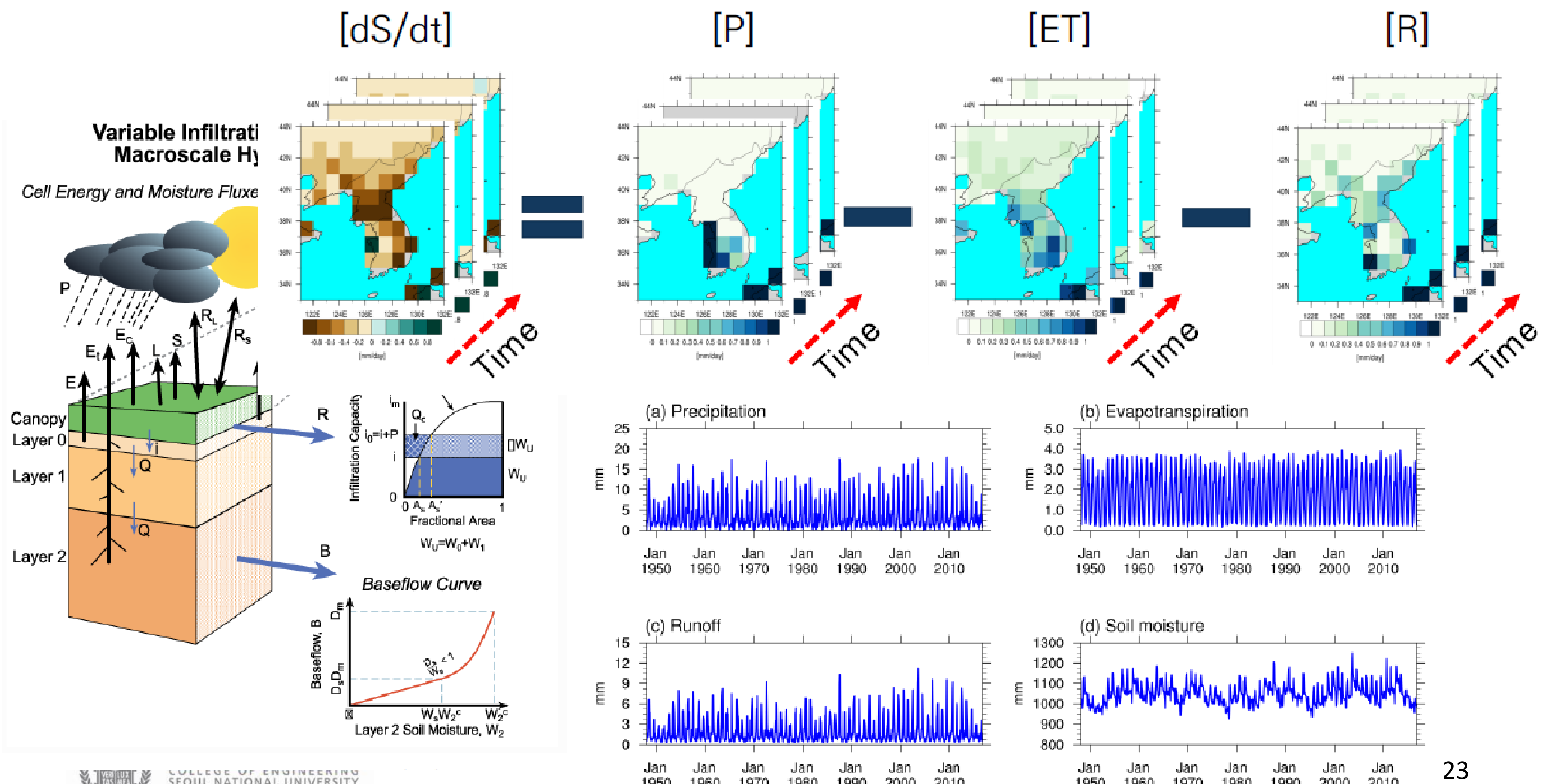


FIG. 2. The SMI satell



Ongoing Research Topics (Co-work)

❖ Relief of water shortage in western Chungcheongnam-do by regenerating inflow and improving water supply adjustment standards



For more,

❖ Related Publications

- C4: **Kim, G. J.**, & Kim, Y.-O.* (2021). How does the coupling of real-world policies with optimization models expand the practicality of solutions in reservoir operation problems? *Water Resources Management*, 35, 3121-3137. https://doi.org/10.1007/s11269-021-02862-y
- C5: **Kim, G. J.**, Seo, S. B., & Kim, Y.-O.* (in press). Adaptive Reservoir Management by Reforming the Zone-based Hedging Rules against Multi-year Droughts. *Water Resources Management.*
- C6: **Kim, G. J.**, Kim, Y.-O.*, & Reed, P. M. (2021). Improving the robustness of reservoir operations with stochastic dynamic programming. *Journal of Water Resources Planning and Management*, *147*(7),04021030. https://doi.org/10.1061/(ASCE)WR.1943-5452.0001381

❖ My personal webpage: https://gijookim.info/

❖ Meet in person in Atlanta, GA (EWRI 2022)

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